

Protecting Margin, Capturing Q4, Closing Revenue Leaks

Executive Decision Report: three strategic recommendations from the retail analytics programme

Prepared for the Executive Committee | October 2026

Based on 3 years of order data (Jan 2023 - Dec 2025). Dataset is synthetic; marketing spend is illustrative.

Executive summary

We grow profit fastest by fixing how we discount and sell, not by selling more everywhere

- Business is stable, not growing fast: 2025 revenue Rs 1.08 Cr (+2.8%), profit Rs 35.5 L at a 32.9% margin.
- Q4 is the engine: October-December earns 44% more per month than the rest of the year, in all three years.
- Electronics is our biggest category (32% of revenue) but earns only 22.6% of profit. Beauty and Fashion earn more profit per rupee.
- Discounts cost margin (38.5% down to 21.6%) and showed no effect on basket size or repeat purchase. All 27 loss-making orders are deep-discounted Electronics.
- Only 81% of orders become revenue: refunds, pending and failed payments hold about Rs 70 L (17.6%) of order value.

Decisions we ask for

1. Approve discount guardrails
2. Approve a profit-weighted Q4 growth plan
3. Approve a payment and refund leakage programme

Combined impact (base case)

About Rs 2.3 L extra annual profit for Rs 1.1 L investment (ROI 107%), payback in about 6 months. Low case roughly breaks even.

Approach and methodology

Four analytical stages, every number traceable to a notebook or the live dashboard

1 Define & protect

KPI dictionary (10 KPIs) and a data-quality contract with 18 checks, thresholds and escalation.

2 Clean

12,420 raw rows to 11,822 trusted orders: duplicates, 5 date formats, currency text, impossible values fixed and flagged.

3 Analyse

EDA with distributions, correlations and outlier review; 3 hypothesis tests with effect sizes and confidence intervals.

4 Monitor

Interactive dashboard: slicers, drill-down, regional map, same KPI engine as this report.

- Revenue = Paid orders only; profit = revenue minus cost of goods (excludes shipping, fees and marketing).
- Seasonality tested on 36 monthly totals (Welch t-test and Mann-Whitney); category margins with Kruskal-Wallis and Tukey; retention with a chi-square cohort test.
- Limits: synthetic data, illustrative marketing spend, observational (not experimental) evidence.

2025 performance snapshot

Revenue and margin are healthy; acquisition economics are the weak spot

REVENUE

Rs 1.08 Cr

+2.8% vs 2024

AVERAGE ORDER VALUE

Rs 3,406

+4.2% vs 2024

ACQUISITION COST (ILLUSTR.)

Rs 2,107

+104.5% vs 2024

MONTHLY CHURN

12.5%

flat vs 2024

PAID ORDERS

3,164

-1.3% vs 2024

GROSS PROFIT

Rs 35.5 L

margin 32.9%

NEW CUSTOMERS

239

-65.8% vs 2024

ACTIVE CUSTOMERS

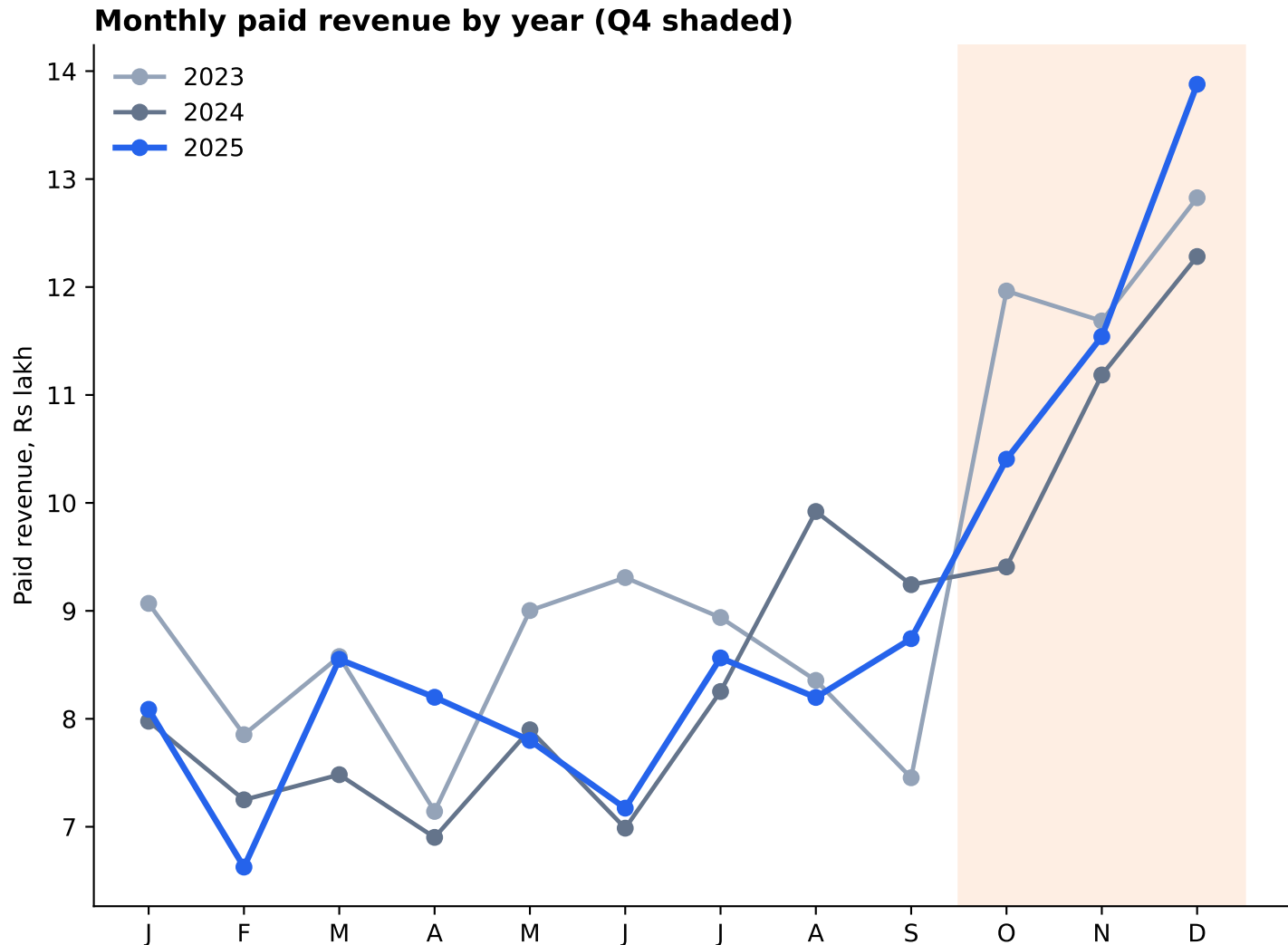
1,317

+0.5% vs 2024

- Revenue grew 2.8% on a 4.2% higher order value while order count slipped 1.3%: growth comes from basket value, not volume.
- New customers fell 66% and CAC doubled: the business increasingly lives off its existing customer base (about 1,300 active).
- Churn is steady at about 12.5% a month, so the base holds only while repeat buying holds.

Revenue driver 1: Q4 is the peak season

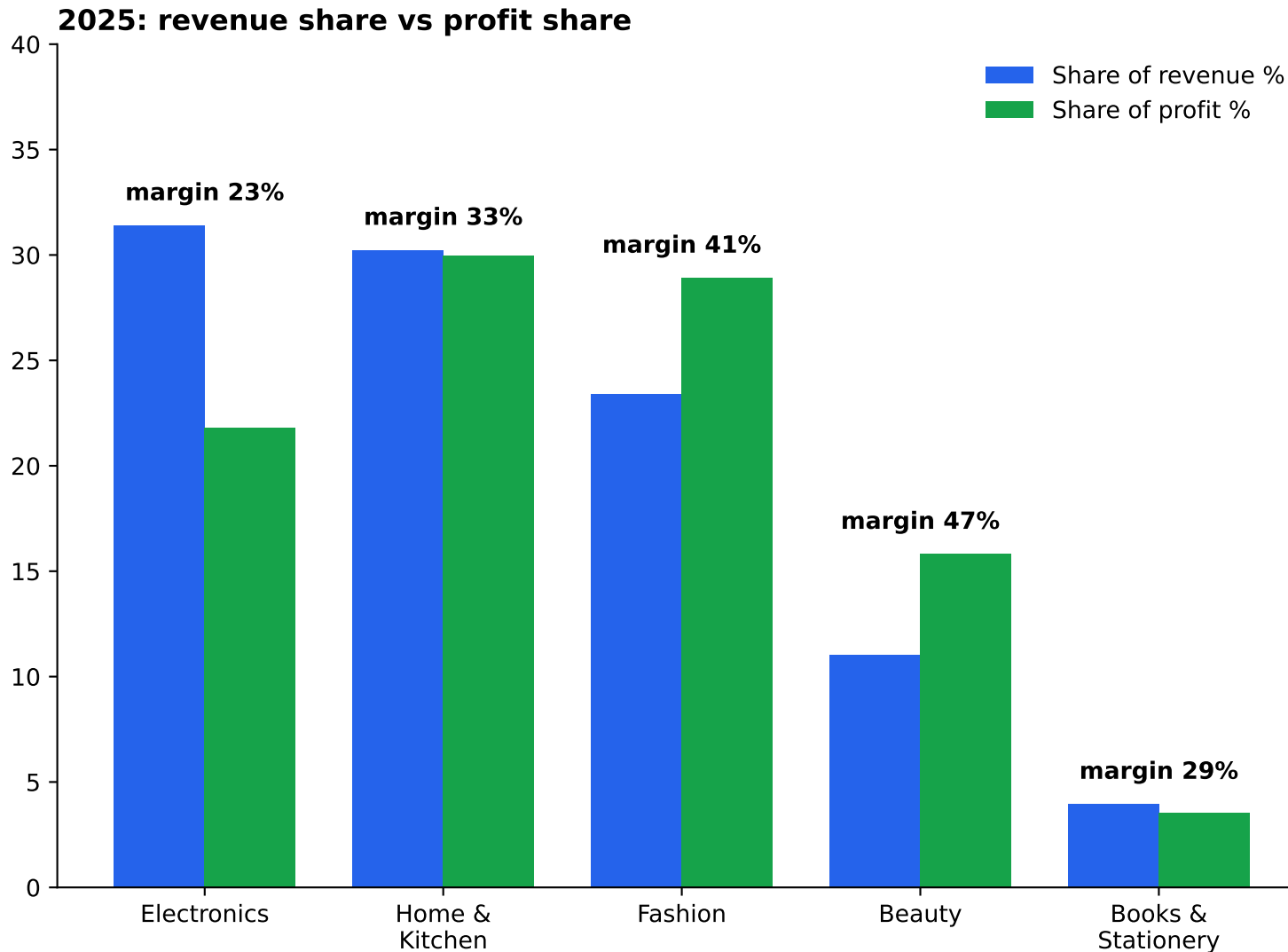
October-December earns 44% more per month, every year



- Q4 months average Rs 11.7 L versus Rs 8.1 L in Jan-Sep: +44% (95% CI +32% to +55%, $p < 0.0001$).
- The uplift holds in each year: +45% (2023), +37% (2024), +49% (2025). It is a stable seasonal pattern.
- Every category peaks in Nov-Dec, so Q4 lifts the whole business.
- So what: stock, logistics, support and ad budget must be ready by August.

Revenue driver 2: category mix decides profit

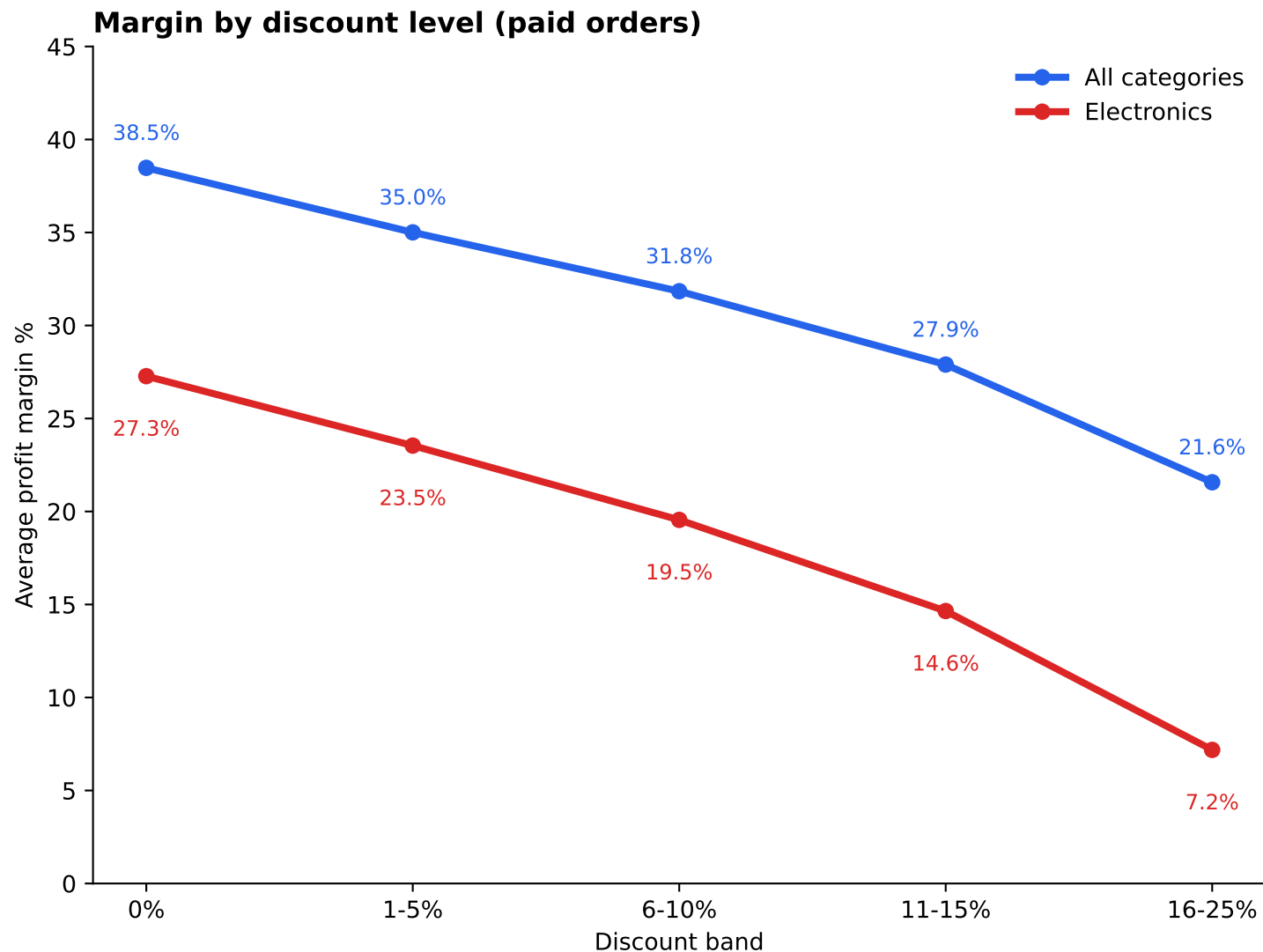
Revenue rank is not profit rank



- Electronics: 32% of revenue, only 22.6% of profit (23% margin).
- Beauty: 10% of revenue but 14.6% of profit (47% margin).
- Category explains about 68% of the variation in margin; all ten category pairs differ significantly.
- So what: judge categories by profit contribution and steer growth spend to Beauty, Fashion and Home & Kitchen.

Risk 1: discounting destroys margin without buying growth

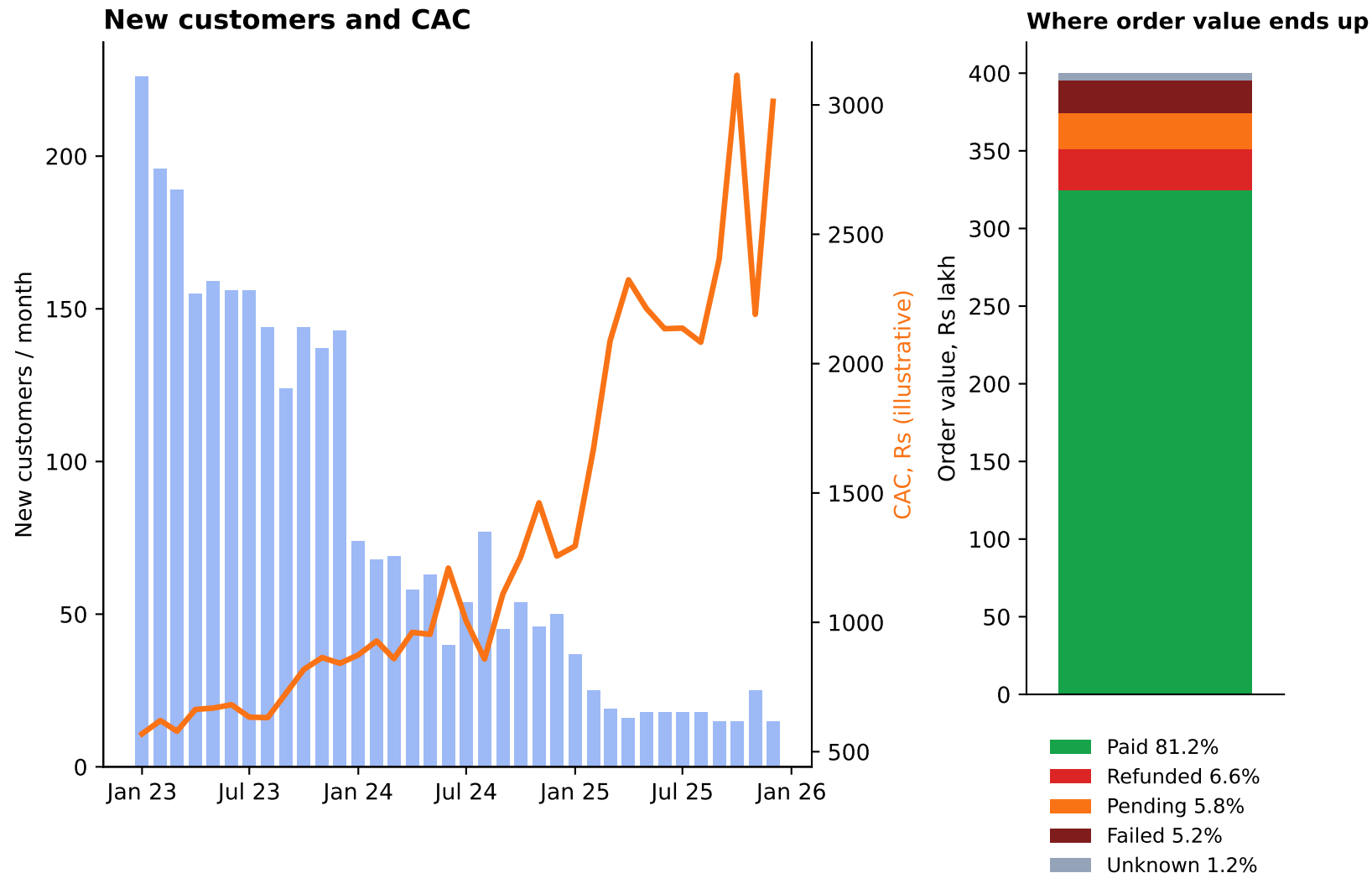
Deeper discount, lower margin, same basket size



- Margin falls from 38.5% (no discount) to 21.6% at 16-25% off.
- Discount has no link to basket size (correlation 0.01) and no effect on 180-day repeat purchase (47.6% vs 47.5%, $p = 0.96$).
- All 27 loss-making orders are Electronics at 20-25% off. Electronics cost is about 72% of list price.
- So what: guardrails first, then test any new discount before scaling.

Risk 2: weaker acquisition and leaking revenue

Illustrative CAC is up 5x since 2023 and nearly 1 in 5 orders never becomes revenue



- New customers fell from 226 a month (Jan 2023) to 15 (Dec 2025). CAC rose from Rs 569 to Rs 3,014.

- Refunded, pending and failed orders hold about Rs 70 L, 17.6% of order value; refund rates are similar across categories (6.4-7.6%), so the cause is process, not product.

- Caveat: spend is illustrative; the customer pool in the data is fixed.

Three strategic recommendations

Protect margin, capture the peak, close the leaks

1 Discount guardrails

Cap Electronics discounts at 15% and add a margin check at promotion setup. Run a controlled test of first-order discounts before any scale-up.

Evidence: margin 38.5% to 21.6%; 27 loss-making orders; no retention effect.

2 Profit-weighted Q4 plan

Pre-build stock and ad budget by August; weight Q4 spend to Beauty, Fashion and Home & Kitchen; track Electronics on margin, not sales.

Evidence: Q4 +44% every year; category margin gap of 24 points.

3 Leakage programme

Reduce failed payments (retries, UPI-first checkout), pending-order follow-up and refund causes (product content, delivery); report a weekly paid-conversion KPI.

Evidence: 81% of orders paid; Rs 70 L of order value not realised.

ROI projections

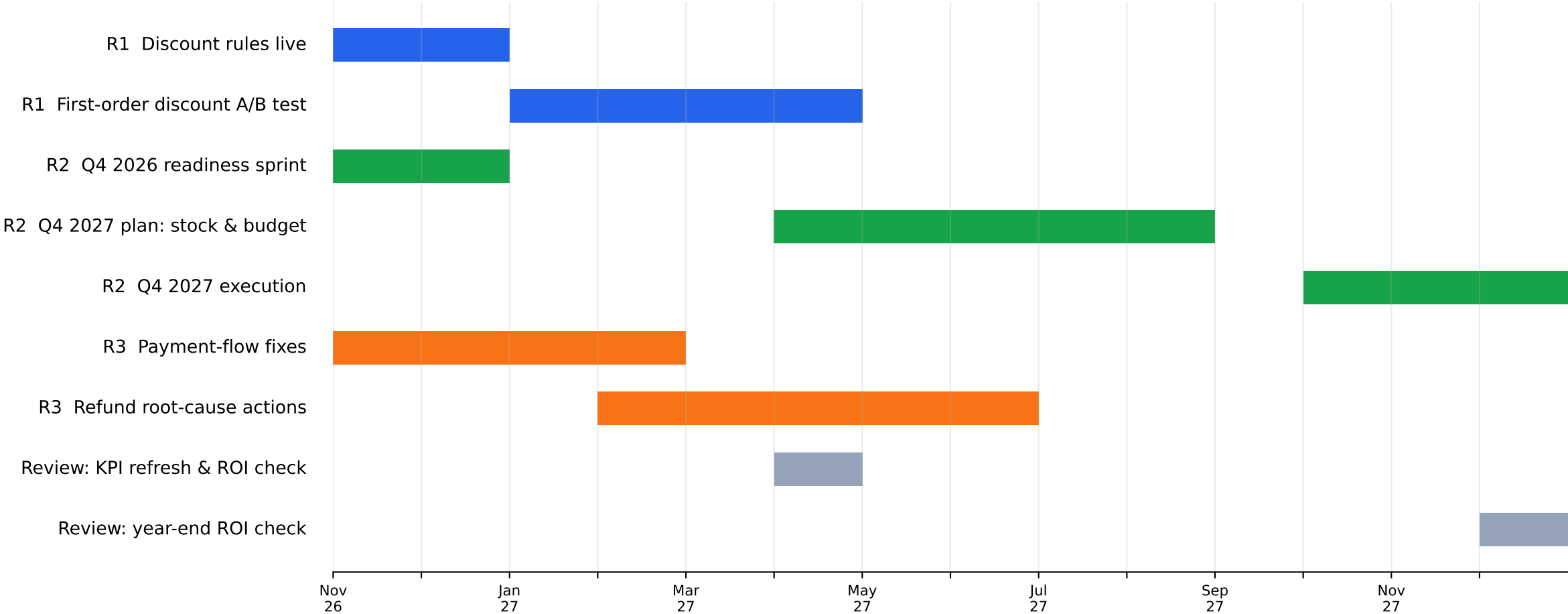
Base case: Rs 2.3 L extra profit a year for Rs 1.1 L of effort; the low case roughly breaks even

Recommendation	Investment	Annual profit: low	Base	High	ROI (base)	Payback	Break-even (% of base)
1 Discount guardrails	Rs 0.10 L	Rs 0.08 L	Rs 0.16 L	Rs 0.24 L	61%	7.5 mo	62%
2 Profit-weighted Q4 plan	Rs 0.40 L	Rs 0.47 L	Rs 0.94 L	Rs 1.42 L	136%	5.1 mo	42%
3 Leakage programme	Rs 0.60 L	Rs 0.59 L	Rs 1.18 L	Rs 1.77 L	96%	6.1 mo	51%
Total	Rs 1.10 L	Rs 1.14 L	Rs 2.28 L	Rs 3.42 L	107%	5.8 mo	48%

- R1: cap Electronics discounts at 15%; 58 orders affected in 2025; assume 15% of them are lost. R2: +8% Q4 revenue at the 2025 margin (32.9%). R3: recover 15% of the Rs 24 L non-paid value at the 2025 margin.
- Low and high cases scale benefits by 0.5x and 1.5x; investment is held fixed. Investment figures are internal-effort planning assumptions for a business of this size (rule configuration, planning and creative re-weighting with no extra media, payment-flow tuning), to be confirmed by Finance. Extra media spend would need to beat roughly 3x return to pay back at a 33% margin.
- Base year is 2025 data; benefits are annual and not discounted. Upside excluded: any first-order discount savings (pending the test).

Implementation timeline

Quick wins first; full Q4 2027 plan in place by September 2027



Owners (proposed): R1 Pricing & Promotions Lead; R2 Head of Growth with Merchandising; R3 Payments Operations & Customer Support.
Governance: monthly KPI review using the dashboard.

Decisions requested and key risks

What we need from the Executive Committee today

Decisions requested

1. Approve the Electronics discount cap and the first-order discount test.
2. Approve the Q4 plan budget and start the Q4 2026 sprint now.
3. Appoint an owner for the payment and refund leakage programme.
4. Commission real marketing-spend data so CAC can be trusted.

Risks and mitigations

Cap reduces Electronics volume: monitor weekly, relax if lost profit exceeds gain.
Q4 over-stocking: stage purchase orders and use last year's curve.
Leakage fixes slower than planned: report conversion weekly.
Evidence is observational: test before scaling.

- Data limits: synthetic dataset; marketing spend illustrative; the data ends Dec 2025, so a refresh is the first action before launch.
- Not covered: shipping and payment fees, competitor pricing, customer-level lifetime value.
- Next analysis: contribution margin after fees; retention drivers beyond discounting; real CAC by channel.

Appendix: analytics portfolio and evidence

Every claim in this deck traces to one of these deliverables

Stage	Deliverable	Files
Task 2	KPI dictionary and data-quality contract	kpi_dictionary.xlsx, data_profile_notebook.ipynb, data_quality_contract.md
Task 3	Data ingestion, cleaning, preprocessing	data_cleaning.ipynb, clean_dataset.csv, raw_retail_sales.csv
Task 4	Exploratory analysis and hypothesis tests	eda_analysis.ipynb
Task 5	Interactive dashboard (Streamlit) and PDF export	app.py, dashboard_metrics.py, Dashboard_Export.pdf
Task 6	Executive decision report (this deck)	Executive_Decision_Report.pdf, build_executive_report.py

Repository: github.com/harinimariyappan-09/data-analytics-project

The live dashboard link is in the repository README (Task 5 section).